The Structural Features of Enhanced Hypercube Networks

Liu Hongmei
College of Science
Three Gorges University
Hubei Yichang 443002, PRC liuhm@ctgu.edu.cn

Abstract

Duo to its topological advantages, the enhanced hypercube network is a very important kind of networks which has huge potential to be widely used in computer networks to improve the performance. To fully realize its potential in those networks where reliability and speed are critical, the topological properties of the enhanced hypercube network need to be further discovered. In this paper, the topological structure of enhanced hypercube networks is analyzed, and consequently the cycles embedded in the enhanced hypercube network have been identified. It has been found that under certain parity conditions, the relationships between the edges and cycles of the enhanced hypercube network can be identified by numbers. The results show that the enhanced hypercube network is both reliable and efficient.

1. Introduction

Interconnection networks are becoming more and more important to modern technologies. In many system designs where interconnection networks are involved, such as distributed system and multiprocessor systems, the selection of network topology is primary important to success, because it is crucial to the speed of inter-node communication. For this reason, the subject of interconnection network topology has attracted a great deal of research in past decades. As the results many successful interconnection network topologies have been proposed, among which the hypercube is an outstanding one. The hypercube has received considerable attention due mainly to its regular structure, small diameter, and good connection with a relatively small node degree ([11]). An n-dimensional hypercube (abbreviated to n-cube) is denoted as Q_n in this paper. Recently many variants of Q_n have been proposed to improve the performance of Q_n, and the most significant ones wielding a lot of influence are generalized hypercube, folded hypercube, twisted hypercube, argument hypercube and enhanced hypercube. ([1], [6], [7], [12], [5], [4], [3], [9]).

As a big effort to improve the efficiency of Q_n, the enhanced hypercubes, denoted by $Q_{n,k}$ $(1 \leq k \leq n-1)$, were proposed ([10]). Inherited from Q_n, $Q_{n,k}$ is also a regular graph. But the enhanced hypercubes are much more attractive than normal hypercubes due to its potential nice topological properties. For instance, in [1], S. A. choudum and R. Usha Nandini showed that a complete binary tree can be embedded in enhanced hypercubes $Q_{n,k}$ under some restricted conditions. Its special case of $k = 1$ is the well-known Folded hypercube (denoted by FQ_n) which has received substantial researches ([7], [12]). It is understood that the hardware cost to construct enhanced hypercubes $Q_{n,k}$ is greater than normal hypercubes Q_n. However, the overhead is negligible when n is large, and the benefit from the structural advantages is overwhelming.

Recent years the research of $Q_{n,k}$ has been fruitful. In [9], the author has obtained that Enhanced hypercube networks $Q_{n,k}$ for $2 \leq k \leq n-1$ have the minimum transmission delay $k - 1 + \lceil \frac{n-k+1}{2} \rceil$. This result also holds for $Q_{n,1}$ which means that the folded hypercube FQ_n has minimum transmission delay $\lceil \frac{n}{2} \rceil$. The author has also obtained that there exist $n+1$ internally-disjoint paths between any two distinct nodes in $Q_{n,k}$ for $k = 2$, and the connectivity $\kappa(Q_{n,2})$ and the edge-connectivity $\kappa'(Q_{n,2})$ are $n+1$. Furthermore, [8] investigated the structural natures of Enhanced hypercube networks in detail and generalized the results of [9] to that there exist $n+1$ internally-disjoint paths between any two distinct nodes in $Q_{n,k}$ for $3 \leq k \leq n-1$, and both the connectivity $\kappa(Q_{n,k})$ and the edge-connectivity $\kappa'(Q_{n,k})$ of $Q_{n,k}$ equal to $n+1$.

Although enhanced hypercube has been proved to have many advantages and displayed its potential applications, a great many of its topological features still remains unknown, which prevents the further applications of enhanced hypercube networks in critical areas. This paper, upon the previous results, aims to further explore these unknown features, particularly those important features already known to Q_n but unknown to $Q_{n,k}$. One of them is the distribution of cycles in $Q_{n,k}$. In [2], the authors proved that every edge of Q_n $(n \geq 2)$ lies on a cycle of every even length from 4 to

978-0-7695-3736-8/09 $25.00 © 2009 IEEE
DOI 10.1109/ICNC.2009.191

Authorized licensed use limited to: Julius-Maximilians-Universitaet Wuerzburg. Downloaded on September 08,2022 at 13:12:52 UTC from IEEE Xplore. Restrictions apply.

2^n. Their results are summarized as the following lemma:

Lemma 1 *([2]) If $n \geq 2$, then*

(a) for any two different vertices x and y in Q_n there exists an $(x, y)-$path of length l with $d_{Q_n}(x, y) \leq l \leq 2^n - 1$, where l and $d_{Q_n}(x, y)$ have same parity;

(b) every edge of Q_n lies on a cycle of every even length from 4 to 2^n.

In this paper, we demonstrate that there is a similar conclusion for $Q_{n,k}$. It is proved that $Q_{n,k}$ is bipartite if and only if n and k have same parity. The minimum length of even cycles in $Q_{n,k}$ is 4, and the minimum length of odd cycles is $n - k + 2$ if n and k have different parity. These results, together with Lemma 1, are then used to show that every edge of $Q_{n,k}$ lies on a cycle of every even length from 4 to 2^n if n and k have same parity, and every edge of $Q_{n,k}$ lies on a cycle of every even length from 4 to 2^n and every odd length of $n - k + 2$ to $2^n - 1$ if n and k have different parity. These excellent structural natures suggest that the enhanced hypercube networks are both reliable and efficient.

2. Preliminaries

Q_n consists of 2^n nodes that are labelled with 2^n binary numbers from 0 to $2^n - 1$. Thus Q_n has vertex set $V(Q_n) = \{x_1x_2 \cdots x_n : x_i = 0, 1. i = 1, 2, \ldots, n\}$, with two vertices $x_1x_2 \cdots x_n$ and $y_1y_2 \cdots y_n$ being adjacent if and only if they differ in exactly one bit.

Suppose x and y are two vertices of Q_n. The Hamming distance of x and y, denoted by $d_H(x, y)$, is the number of different bits between the corresponding strings of x and y. Let $d_{Q_n}(x, y)$ denote the distance between x and y, i.e., the length of the distance of the shortest (x, y)-path in Q_n. Clearly, $d_H(x, y) = d_{Q_n}(x, y)$. For any $x_i \in \{0, 1\}$, the complementation of $x_i \in \{0, 1\}$ is denoted by $\overline{x}_i$.

A vertex x in Q_n is said to be odd or even if the sum of its bits is odd or even. Let $X = \{x : x \text{ is odd}\}$ and $Y = \{x : x \text{ is even}\}$. Then $\{X, Y\}$ is a bipartition of Q_n. Conversely, if two vertices of Q_n are in the same part of a bipartition of Q_n, then they have same parity.

As mentioned above, the folded hypercube is a typical network topology vastly used in practice. It is a special kind of enhanced hypercubes when $k = 1$. Thus enhanced hypercube can be regarded as a generalization of folded hypercube. Furthermore, Hypercube is a subgraph of enhanced hypercube obtained by removing edges of $(x_1x_2 \cdots x_{k-1}x_k \ldots x_n,\ x_1x_2 \cdots x_{k-1}\ \overline{x}_k\overline{x}_{k+1}\ \ldots \overline{x}_n)$. These removed edges are called compliment edges of $Q_{n,k}$ from the enhanced hypercube. The first kind of edges are called the edges of Q_n.

From the definition of $Q_{n,k}$, it is easy to derive the following properties:

Lemma 2 *Enhanced hypercube $Q_{n,k}$ $(1 \leq k \leq n - 1)$ is a graph. The set of the vertices is denoted as $V = \{(x_1x_2 \cdots x_n) : x_i = 0, 1. 1 \leq i \leq n\}$. Two vertices x and y are adjacent if y satisfies one of the following two conditions:*

1. $(x_1x_2 \cdots x_{i-1}\overline{x}_ix_{i+1} \ldots x_n)$, $1 \leq i \leq n$, or

2. $(x_1x_2 \cdots x_{k-1}\ \overline{x}_k\overline{x}_{k+1}\ \ldots \overline{x}_n)$.

Lemma 3 *$Q_{n,k}$ is $(n + 1)$-regular and has 2^n vertices and $(n + 1)2^{n-1}$ edges.*

Lemma 4 *$Q_{n,k}$ can be decomposed into two subsystems with respect of some components i $(1 \leq i \leq n)$ such that $Q_{n,k} = Q^{i0}_{n,k} \bigcup Q^{i1}_{n,k}$. Node $x_1x_2 \cdots x_n \in V(Q^{i0}_{n,k})$ if and only if the ith component $x_i = 0$; $x_1x_2 \cdots x_n \in V(Q^{i1}_{n,k})$ if and only if its ith component $x_i = 1$. If $i < k$, $Q^{i0}_{n,k}$ and $Q^{i1}_{n,k}$ are two $(n-1)$-dimensional enhanced hypercube, the bound $[V(Q^{i0}_{n,k}),\ V(Q^{i1}_{n,k})]$ consists of some edges of Q_n; If $i > k$, $Q^{i0}_{n,k}$ and $Q^{i1}_{n,k}$ are two $(n - 1)$-dimensional hypercube Q_{n-1}, the bound $[V(Q^{i0}_{n,k}),\ V(Q^{i1}_{n,k})]$ consists of all compliment edges of $Q_{n,k}$.*

3. The Structural Properties of $Q_{n,k}$

This section is used to analyse the topological structure of $Q_{n,k}$. Based on the analysis, we will obtain our main results regarding the distribution of cycles in $Q_{n,k}$.

Theorem 1 *$Q_{n,k}$ is a bipartite graph if and only if n and k have same parity.*

Proof Since $Q_{n,k}$ is obtained from Q_n by adding 2^{n-1} complementary edges, to prove the theorem, it is sufficient to consider complementary edges. Suppose that $x = x_1x_2 \cdots x_n$, $\overline{x} = x_1x_2 \cdots x_{k-1}\overline{x}_k\overline{x}_{k+1} \ldots \overline{x}_n$, then $x\overline{x}$ is a complementary edge of $Q_{n,k}$.

Case 1. Let n is even.

If k is even, then $n - k + 1$ is odd, the vertex x and its complementary $\overline{x}$ have different parity;

If k is odd, then $n - k + 1$ is even, the vertices x and $\overline{x}$ have same parity.

Case 2. Let n is odd.

If k is even, then $n - k + 1$ is even, the vertices x and $\overline{x}$ have same parity;

If k is odd, then $n - k + 1$ is odd, the vertices x and $\overline{x}$ have different parity.

Therefore x and its complementary $\overline{x}$ in $Q_{n,k}$ have different parity if and only if n and k have same parity. Any complementary edge in $Q_{n,k}$ joins two vertices from different parts of $\{X, Y\}$. It follows that $Q_{n,k}$ is bipartite if and only if n and k have same parity. The conclusion of the theorem follows.

Theorem 1 is useful when the parity of the length of cycles in $Q_{n,k}$ needs to be decided, as shown in the following theorem.

Authorized licensed use limited to: Julius-Maximilians-Universitaet Wuerzburg. Downloaded on September 08,2022 at 13:12:52 UTC from IEEE Xplore. Restrictions apply.

Theorem 2 *(1) The length of any cycle in $Q_{n,k}$ containing exactly two complementary edges is even. (2) The length of any cycle in $Q_{n,k}$ containing exactly one complementary edge is at least $n-k+2$.*

Proof (1) Let $C = v\overline{v}v_1v_2\dots v_m\overline{u}uu_1\dots u_nv$ be a cycle in $Q_{n,k}$ that contains exactly two complementary edges $e_1 = u\overline{u}$ and $e_2 = v\overline{v}$. Clearly, $u \neq v$, and hence $\overline{u} \neq \overline{v}$ because $e_1 \neq e_2$.

If $Q_{n,k}$ is bipartite, then we are done. Now assume that $Q_{n,k}$ is not bipartite. Since Q_n is bipartite, u and $\overline{u}$ have same parity. Similarly v and $\overline{v}$ have same parity. It is known that C contains exactly two complementary edges, and all of other edges in C are in Q_n. If u, $\overline{u}$, v, $\overline{v}$ are in the same part of $\{X,\ Y\}$, then the lengths of the sections $\overline{v}v_1v_2\dots v_m\overline{u}$ and $uu_1\dots u_nv$ are even. If u, $\overline{u}$, v, $\overline{v}$ are not in the same part, then the lengths of the two sections are odd. It means that the length of C is even in both cases.

(2) Now assume that $e = u\overline{u}$ is a complementary edge of $Q_{n,k}$ and C is a cycle containing exactly one complementary edge e. Then $P = C - e$ is a $(u,\overline{u})-$path in Q_n. Since $d_{Q_n}(u,\overline{u}) = n-k+1$ and the length of any $(u,\overline{u})-$path in Q_n is at least $n-k+1$, the length of P is at least $n-k+1$ and the length of the cycle C is at least $n-k+2$. The proof has been completed.

Shortest paths play an extremely important role in the routings in computer networks. The following theorem aims to discover the features for the shortest paths of $Q_{n,k}$.

Theorem 3 *(1) Every shortest path between any two distinct vertices in $Q_{n,k}$ contains at most one complementary edge. (2) Suppose $x = x_1x_2\dots x_n$ and $y = y_1y_2\dots y_n$ are two distinct vertices in $Q_{n,k}$. Any shortest $(x,y)-$path in $Q_{n,k}$ contains no complementary edges if and only if $\sum_{i=k}^{n}|x_i - y_i| \leq \frac{n-k}{2}+1$. (3) If $Q_{n,k}$ contains an odd cycle, then any shortest odd cycle contains exactly one complementary edge and the length is $n-k+2$.*

Proof (1) Suppose that $P = uu_1\dots u_i\overline{u}_iu_{i+1}\dots u_j\overline{u}_j\dots v$ is a shortest $(u,v)-$path between two vertices u and v of $Q_{n,k}$ containing more than one complementary edge, where $u_i\overline{u}_i$ and $u_j\overline{u}_j$ are two complementary edges that first occur in P in the order from u to v. Then the section $\overline{u}_iu_{i+1}\dots u_j$ of P contains no complementary edge and is a shortest $(\overline{u}_i,u_j)-$path, therefore P is not the shortest path, a contradiction.

(2) For the sake of convenience, let $l = \sum_{i=1}^{k-1}|x_i - y_i|$, $r = \sum_{i=k}^{n}|x_i - y_i|$ and $y^* = y_1y_2\dots y_{k-1}x_k\dots x_n$. There are two paths P and Q joining x and y in $Q_{n,k}$.

Path P can be constructed as following:

$$\begin{aligned} x \quad &\to y_1x_2\dots x_n \to y_1y_2x_3\dots x_n \to \dots \\ &\to y_1y_2\dots y_{k-1}x_k\dots x_n \to y_1y_2\dots y_kx_{k+1}\dots x_n \\ &\to \dots \to y = y_1y_2\dots y_n \end{aligned}$$

where P contains no complementary edges, and $d_P(x,y) = l+r$.

Path Q can be constructed as following:

$$\begin{aligned} x \quad &\to y_1x_2\dots x_n \to y_1y_2x_3\dots x_n \to \dots \\ &\to y_1\dots y_{k-1}x_k\dots x_n \\ &\to y_1\dots y_{k-1}\overline{x}_k\overline{x}_{k+1}\dots\overline{x}_n \\ &\to y_1\dots y_{k-1}y_k\overline{x}_{k+1}\dots\overline{x}_n \\ &\to y_1\dots y_{k-1}y_ky_{k+1}\overline{x}_{k+2}\dots\overline{x}_n \to \dots \to y \end{aligned}$$

where Q contains exactly one complementary edge $y*\overline{y}*$, and $d_Q(x,y) = l+n-k-r+2$. Any shortest path connecting x and y contains no complementary edge if and only if $d_H(x,y) = d_{Q_n}(x,y) = l+r$ and also if and only if $r \leq \frac{n-k}{2}+1$.

(3) If $Q_{n,k}$ contains an odd cycles, then $Q_{n,k}$ is not bipartite. Suppose that $C = x_0x_1\dots x_{2m}x_0$ is a shortest odd cycle of length 2m+1 in $Q_{n,k}$. Since any complementary edges are not adjacent, it follows that C contains at least one non-complementary edge, say $e = v_mv_{m+1}$, which is an edge of Q_n. Consider two parts of C, $P = v_0v_1\dots v_m$ and $Q = v_{m+1}\dots v_{2m}v_0$. Theorem 3 confirms that P and Q contains at most one complementary edge respectively. If both of them contain one complementary edge, then C contains two complementary edges. Hence, with Theorem 2, C is an even cycle. However, C is odd, so C contains exactly one complementary edge.

Without loss of generality, assume that $e = x_lx_{l+1} = x_l\overline{x}_l$ is a complementary edge. Then the shortest $(x_l,\overline{x}_l)-$path T in Q_n is $n-k+1$, that is, $d_H(x_l,\overline{x}_l)$=$d_{Q_n}(x_l,\overline{x}_l)$=$n-k+1$. Therefore $C = T\bigcup x_l\overline{x}_l$ has length of $n-k+2$. The proof is finished.

Theorem 4 *Every complementary edge in $Q_{n,k}$ lies on a cycle of every even length from 4 to 2^n for $n \geq 2$.*

Proof Let $e = x\overline{x}$ be a complementary edge of $Q_{n,k}$. Without loss of generality, assume that $Q_{n,k} = Q_{n,k}^{k0}\bigcup Q_{n,k}^{k1}$ and $x \in V(Q_{n,k}^{k0})$, $\overline{x} \in V(Q_{n,k}^{k1})$, namely, $x = x_1\dots x_{k-1}0x_{k+1}\dots x_n$, $\overline{x} = x_1\dots x_{k-1}1\overline{x}_{k+1}\dots\overline{x}_n$. Then $[V(Q_{n,k}^{k0},\ V(Q_{n,k}^{k1})]$ consists of all complementary edges. Select a complementary edge $y\overline{y}$ such that $y \in V(Q_{n,k}^{k0})$, $\overline{y} \in V(Q_{n,k}^{k1})$ and then xy, $\overline{xy}$ are edges of $Q_{n,k}$. We now prove that $e = x\overline{x}$ lies on a cycle of length l in $Q_{n,k}$ with l being even and $4 \leq l \leq 2^n$.

Let $l = l'+l''+2$, where $1 \leq l',\ l'' \leq 2^{n-1}-1$ and both l', l'' have same parity with $d_{Q_n}(x,y) = 1$. By Lemma 1(a), there exists a path P of length l' in $Q_{n,k}^{k0}$ joining x and y and a path Q of length l'' in $Q_{n,k}^{k0}$ joining $\overline{x}$ and $\overline{y}$. Then $P\bigcup x\overline{x}\bigcup Q\bigcup y\overline{y}$ is a cycle containing the edge e in $Q_{n,k}$, whose length is l. The proof is finished.

The following main result describes the distribution of cycles in $Q_{n,k}$.

Authorized licensed use limited to: Julius-Maximilians-Universitaet Wuerzburg. Downloaded on September 08,2022 at 13:12:52 UTC from IEEE Xplore. Restrictions apply.

Theorem 5 *If n and k have same parity, then every edge of $Q_{n,k}$ lies on a cycle of every even length from 4 to 2^n. If n and k have different parity, then every edge of $Q_{n,k}$ lies on a cycle of every even length from 4 to 2^n and every odd length of $n-k+2$ to 2^n-1.*

Proof If n and k have same parity, then $Q_{n,k}$ is a bipartite graph by Theorem 1. According to Lemma 1(b) and Theorem 4, the first assertion holds.

If n and k have different parity, then $Q_{n,k}$ is not a bipartite graph. By Lemma 1(b) and Theorem 4, every edge lies on a cycle of every even length from 4 to 2^n.

Next, it will be proved that every edge of $Q_{n,k}$ lies on a cycle of any odd length l for $n-k+2 \leq l \leq 2^n-1$.

Let e be any edge of $Q_{n,k}$. There are two cases to be considered.

Case 1. The edge $e = x\overline{x}$ is a complementary. In this case, $d_{Q_n}(x,\overline{x}) = n-k+1$, which is even. By Lemma 1(a), there is a path P of even length $m,\ n-k+1 \leq m \leq 2^n-2$. Then $P \bigcup x\overline{x}$ is a cycle containing e with length of $l = m+1,\ n-k+2 \leq l \leq 2^n-1$, and l is odd.

Case 2. The edge $e = xy$ is not a complementary edge. Without loss of generality, suppose that $x = x_1x_2\ldots x_{k-1}0x_{k+1}\ldots x_n,\ x \in V(Q^{k0}_{n,k})$, and $y = x_1x_2\ldots x_{k-1}1x_{k+1}\ldots x_n,\ y \in V(Q^{k1}_{n,k})$. Then $\overline{y} = x_1x_2\ldots x_{k-1}0\overline{x}_{k+1}\ldots\overline{x}_n,\ \overline{y} \in V(Q^{k0}_{n,k})$. Since $d_{Q_n}(x,\overline{y}) = n-k$ is odd, by Lemma 1(a), there exists a path P in $Q^{k0}_{n,k}$ with length of odd $m,\ n-k \leq m \leq 2^{n-1}-1$. Therefore, $P \bigcup xy \bigcup y\overline{y}$ is an odd cycle containing edge e in $Q_{n,k}$ with length of $l = m+2$, $n-k+2 \leq l \leq 2^{n-1}+1$.

When $2^{n-1}+3 \leq l \leq 2^n-1$, let $l' = l-2^{n-1}-1$. It follows that $2 \leq l' \leq 2^{n-1}-2$. Suppose that P' is a path with length of $2^{n-1}-1$ connecting x and $\overline{y}$ in $Q^{k0}_{n,k}$. Let $P' = xu_1u_2\ldots u_{2^{n-1}-2}\overline{y}$. Then there is a corresponding path $P'' = yu'_1u'_2\ldots u'_{2^{n-1}-2}\overline{y}'$ in $Q^{k1}_{n,k}$. That is to say, for every vertex $u \in P'$, there is a vertex $u' \in P''$ such that u and u' differ only in the kth bit. Consequently we are able to construct an odd cycle $C = xu_1u'_1u'_2u_2u_3u'_3\ldots u'_lu_lu_{l+1}\ldots u_{2^{n-1}-2}\overline{y}yx$ with length of $2^{n-1}-1+l'+2 = l$ in $Q_{n,k}$.

The theorem is proved.

4. Comments and Conclusion

Upon the topological structure analysis of $Q_{n,k}$, this paper identified a number of novel features, which then have been used to prove that $Q_{n,k}$ is bipartite if and only if n and k have same parity and the minimum length of even cycles in $Q_{n,k}$ is 4, and the minimum length of odd cycles is $n-k+2$ if n and k have different parity. It is also proved that if n and k have same parity, then every edge of $Q_{n,k}$ lies on a cycle of every even length from 4 to 2^n, and if n and k have different parity, then every edge of $Q_{n,k}$ lies on a cycle of every even length from 4 to 2^n and every odd length of $n-k+2$ to 2^n-1. These good properties imply that the reliability, efficiency and fault tolerance of $Q_{n,k}$ are better than hypercube Q_n, which shows that the enhanced hypercube is an excellent choice of network topology for parallel processing computer systems. It is then concluded that interconnection networks modelled by enhanced hypercube are extremely robust.

Acknowledgment

This project is supported by NSFC (10671081).

References

[1] S. A. choudum and N. R. Usha. Complete binary trees in folded and enhanced cubes. *Networks*, 43(4):266–272, 2004.

[2] L. K. Li, C. H. Tsai, J. M. Tan, and L. H. Hsu. Bipanconnectivity and edge-fault-tolerant bipancyclicity of hypercubes. *Inform. Process. lett.*, 87:107–110, 2003.

[3] H. Liu. The restricted connectivity and restricted fault diameter in m-ary n-cube systems. *The 3rd international conference on impulsive dynamic systems and applications*, 4:1368–1371, July 2006.

[4] H. Liu. The routing algorithm for generalized hypercube. *Mathematics in Practice and Theory*, 36(5):258–261, 2006.

[5] H. Liu. Topological properties for m-ary n-cube. *Journal of Wuhan University of Technology (Transportation Science and Engineering)*, 30(2):340–343, 2006.

[6] H. Liu. The pancyclity of m-ary n-cube. *Proceedings of the 2007 IEEE International conference on networking ,sensing and control*, pages 614–617, April 2007.

[7] H. Liu. The performance guaranteed new algorithm for fault-tolerant routing in folded cubes. *Frontiers in Algorithmcs, Lecture Notes on Computer Science*, pages 237–243, August 2007.

[8] H. Liu. Properties and performance of enhanced hypercube networks. *The 4th International Conference on Networked Computing and Advances in Information Management*, pages 280–283, 5 2008.

[9] H. Liu. Properties of enhanced hypercube networks. *Journalof Systems Science and information*, 6(3):251–256, 12 2008.

[10] N. F. Tzeng and S. Wei. Enhanced hypercube. *IEEE Transct. Compnt.*, 40:284–294, 1991.

[11] J. Xu. *Topological structure and Analysis of interconnection networks*. Kluwer Academic Publishers, Dordrecht, Boston, Lodon, 2001.

[12] J. Xu and M. Ma. Cycles in folded hypercubes. *Applied Mathematics Letters*, 19:140–145, 2006.

Authorized licensed use limited to: Julius-Maximilians-Universitaet Wuerzburg. Downloaded on September 08,2022 at 13:12:52 UTC from IEEE Xplore. Restrictions apply.